

Integrating Computer Vision and Dynamic Control for Environmental Perception and Navigation

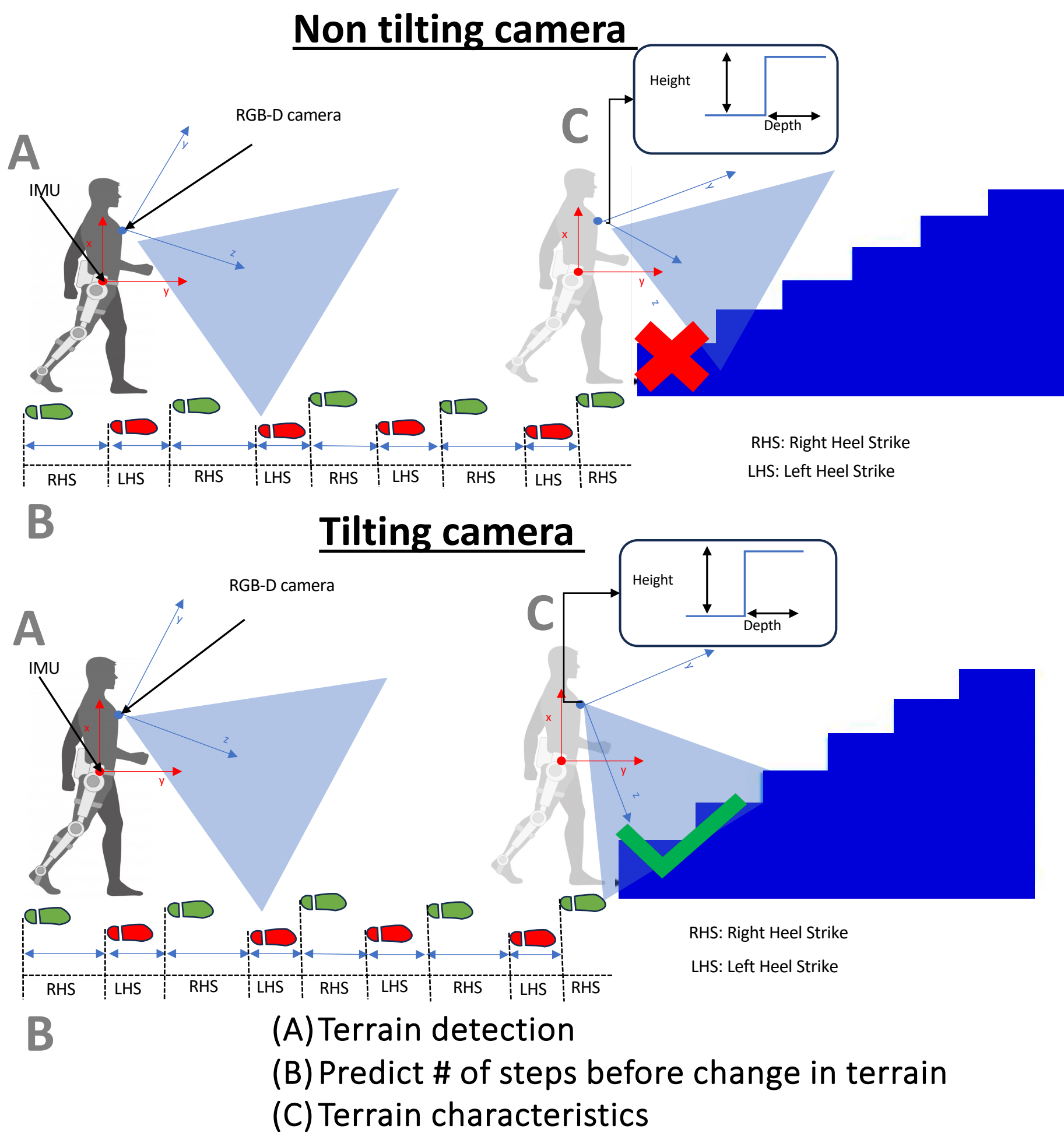
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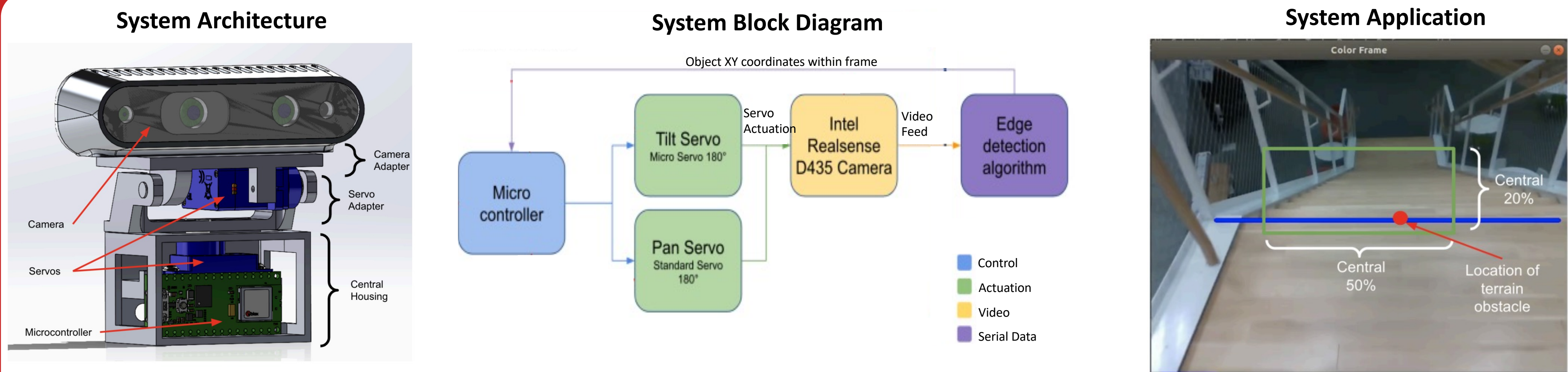
Abstract

This study introduces a novel mechatronic system that integrates computer vision and dynamic control to optimize lower limb prosthetics and exoskeleton support across various terrains. This work introduces a motorized chest mount camera mount that dynamically adjusts to maintain an optimal field of view (FOV). This setup ensures that terrain features crucial for navigation are continuously centered, achieving an accuracy rate of 83%. Our framework employs an algorithmic suite for terrain classification, step measurement, and staircase detection. Using the YOLOv3 model, it detects ascending stairs with 98.1% accuracy, while an edge detection approach locates descending stair navigation with minimal errors. An integrated Inertial Measurement Unit (IMU) and RGB-D camera precisely capture step lengths, facilitating accurate predictions of the transitional step with 95% accuracy. Furthermore, point cloud analysis is utilized to measure staircase dimensions with a high degree of accuracy, demonstrating an average absolute error of 0.84 cm. This integration of advanced sensing, processing, and control technologies significantly enhances the functionality and adaptability of exoskeletons, paving the way for more effective and versatile assistive devices.

Overview



Methods

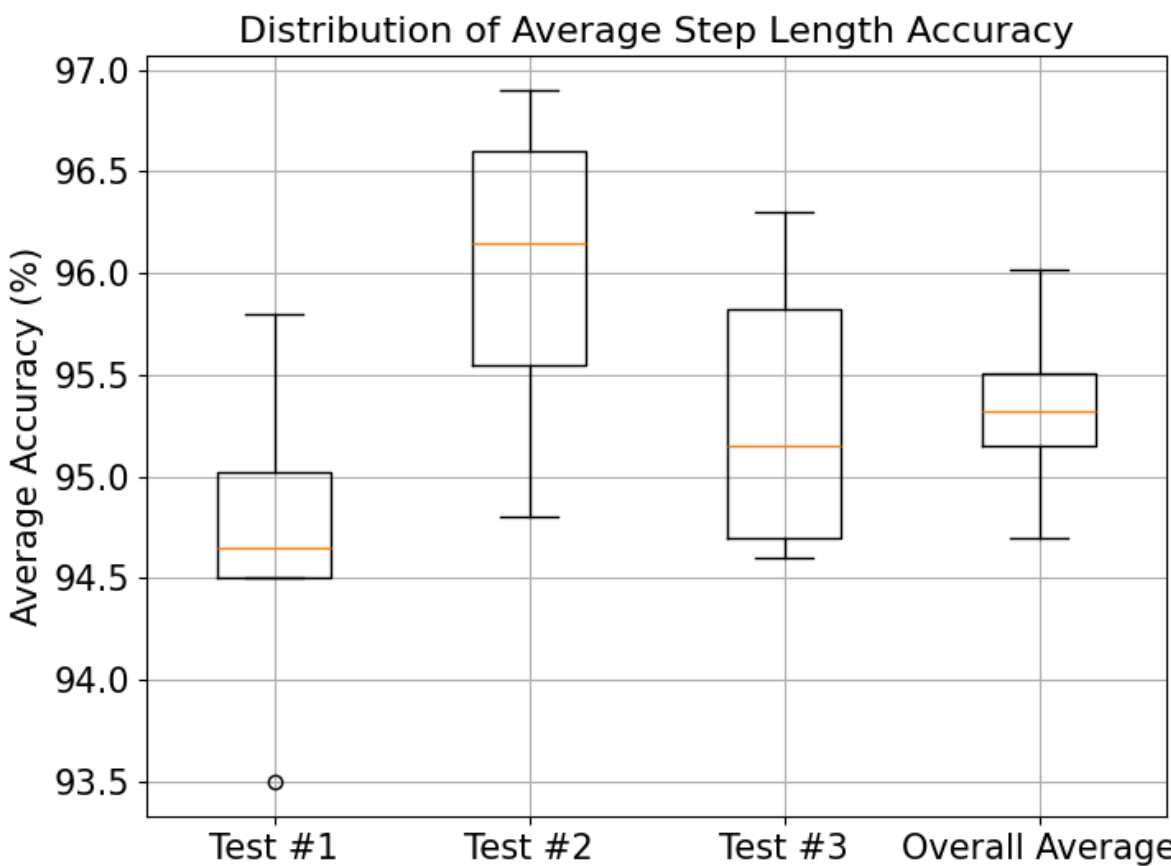


Results

A. Terrain Detection

The network classification accuracy based on the number of true positives during real-time performances reached an average of 98.1% precision score.

B. Step Length Accuracy



C. Terrain Characteristics

Staircase #	Ground Truth Height	Estimated Height Upstairs/Downstairs	Abs Error Difference for Height Upstairs/Downstairs	Ground Truth Depth	Estimated Depth upstairs/Downstairs	Abs Error Difference for Depth Upstairs/Downstairs
1	17.5cm	18.49cm/19.52cm	0.99cm/2.02cm	30cm	30.58cm/28.96cm	0.58cm/1.04cm
2	18cm	17.26cm/20.02cm	0.74cm/2.02cm	28cm	27.16cm/28.76cm	0.84cm/0.76cm
3	16.5cm	16.33cm/18.89cm	0.17cm/2.39cm	30cm	29.18cm/28.94cm	0.82cm/1.07cm

D. System Validation Metrics

Requirement	Specification	Actual
Wearability	Weight	0.285 kg
Wearability	Donning Time	10.51 sec
Power	Current Draw	0.625A
Performance	Effective FOV Increase	162.3°
Accuracy	System Success Rate	83%